Diksha Utekar **D15A-63/64**
Devansh Wadhwani

WT Exp 5

Aim :- To run a simulation of the project on TinkerCAD.

Theory :-
Tinkercad is a free, web-based 3D modeling and design application that has gained significant popularity among beginners, educators, and hobbyists. Developed by Autodesk, Tinkercad simplifies the process of 3D design, making it accessible to users of all skill levels. Since its launch in 2011, it has become a go-to platform for creating models for 3D printing, electronics prototyping, and coding.

Overview of Tinkercad

Type of Site : 3D modeling, Computer Aided Design (CAD)

Owner : Autodesk

URL : www.tinkercad.com

Commercial : Yes, but the software is free to use

Registration : Required to access full features

Launched : 2011

Written in : WebGL, JavaScript

Key Features and Uses

- **3D Design and Printing** :
  Tinkercad is best known for its 3D design capabilities. Users can create detailed 3D models by combining basic shapes such as cubes, spheres, and cylinders. The intuitive interface allows users to drag and drop these shapes onto a work plane and manipulate them to build complex designs.

- **Electronics Prototyping :**
  In addition to 3D design, Tinkercad supports electronics prototyping. Users can create and simulate circuits using a variety of components, including microcontrollers, sensors, and LEDs.

- **Coding :**
  Tinkercad also includes a block-based coding environment, which is ideal for beginners learning to code.

- **User-Friendly Interface :**
  The simplicity of the interface makes it an excellent choice for students, educators, and hobbyists who are new to 3D modeling and design

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- **Accessibility :**
  Since Tinkercad is a browser-based application, it can be accessed from any device with an internet connection. This makes it highly versatile and convenient for users who need to work on their designs from different locations

- **Educational Applications :**
  Tinkercad has become a popular tool in educational settings. It is used by teachers to introduce students to the concepts of 3D design, electronics, and coding.

Code :-

```
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 2, 16);
int sensorPin = A0;
void setup()
{
  Serial.begin(9600);
  // initialize LCD
  lcd.init();
  // turn on LCD backlight
  lcd.backlight();
}
void loop() {
  int sensorValue = analogRead(sensorPin);
  Serial.println(sensorValue);
  int turbidity = map(sensorValue, 0, 520, 100, 0);
  delay(100);
  lcd.setCursor(0, 0);
  lcd.print("Turbidity:");
  lcd.print("    ");
  lcd.setCursor(10, 0);
  lcd.print(turbidity);

  delay(100);
  if (turbidity < 20) {
    lcd.setCursor(0, 1);
    lcd.print(" its CLEAR ");
    Serial.print(" its CLEAR ");
  }
  if ((turbidity > 20) && (turbidity < 50)) {
    lcd.setCursor(0, 1);
    lcd.print(" its CLOUDY ");
    Serial.print(" its CLOUDY ");
  }
```

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Steps:

- To connect a turbidity sensor to an Arduino the wiring is straightforward. Most turbidity sensors (like the SEN0189 from DFRobot) come with 3 or 4 pins. Here's the standard pin connection:

Turbidity Sensor Pinout and Arduino Connection:

- **VCC** on the turbidity sensor connects to the **5V** pin on the Arduino.
- **GND** on the sensor connects to **GND** on the Arduino.
- **OUT** (analog output) on the sensor connects to **A0** on the Arduino.

Example Code to Read Analog Turbidity Value:

```
int turbidityPin = A0;
int turbidityValue = 0;

void setup() {
Serial.begin(9600);
}

void loop() {   turbidityValue =
analogRead(turbidityPin);
  Serial.print("Turbidity Value: ");
Serial.println(turbidityValue);   delay(1000);
}
```

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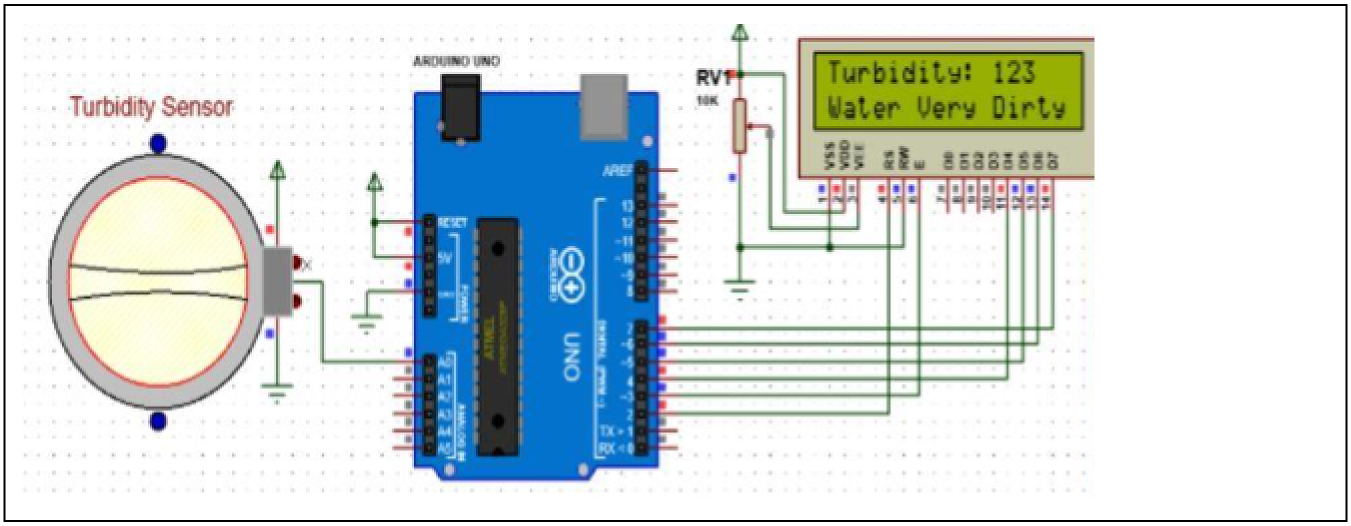
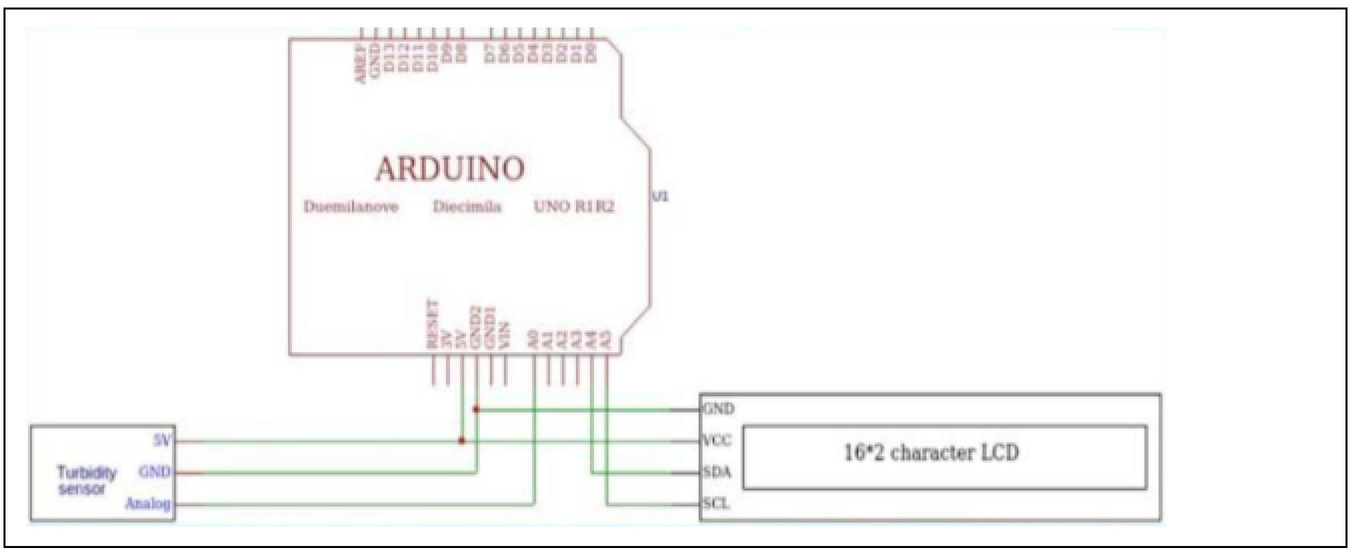
Circuit diagram :



Schematic obtained from TinkerCAD :



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Conclusion :-

The turbidity monitoring system developed using Arduino successfully measures the clarity of water by detecting the level of suspended particles. By interfacing the turbidity sensor with Arduino, real-time data can be obtained and analyzed to determine water quality. The analog values received from the sensor provide a reliable indication of turbidity, which can be used for environmental monitoring, water treatment facilities, or smart IoT-based applications.

This project demonstrates the feasibility of low-cost and efficient water quality monitoring solutions. With further enhancements such as cloud integration, threshold-based alerts, and automated responses, the system can be scaled for broader applications in smart city and environmental conservation projects.